

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Machine Learning Enabled Predictive Analytics for Diabetes Risk Detection

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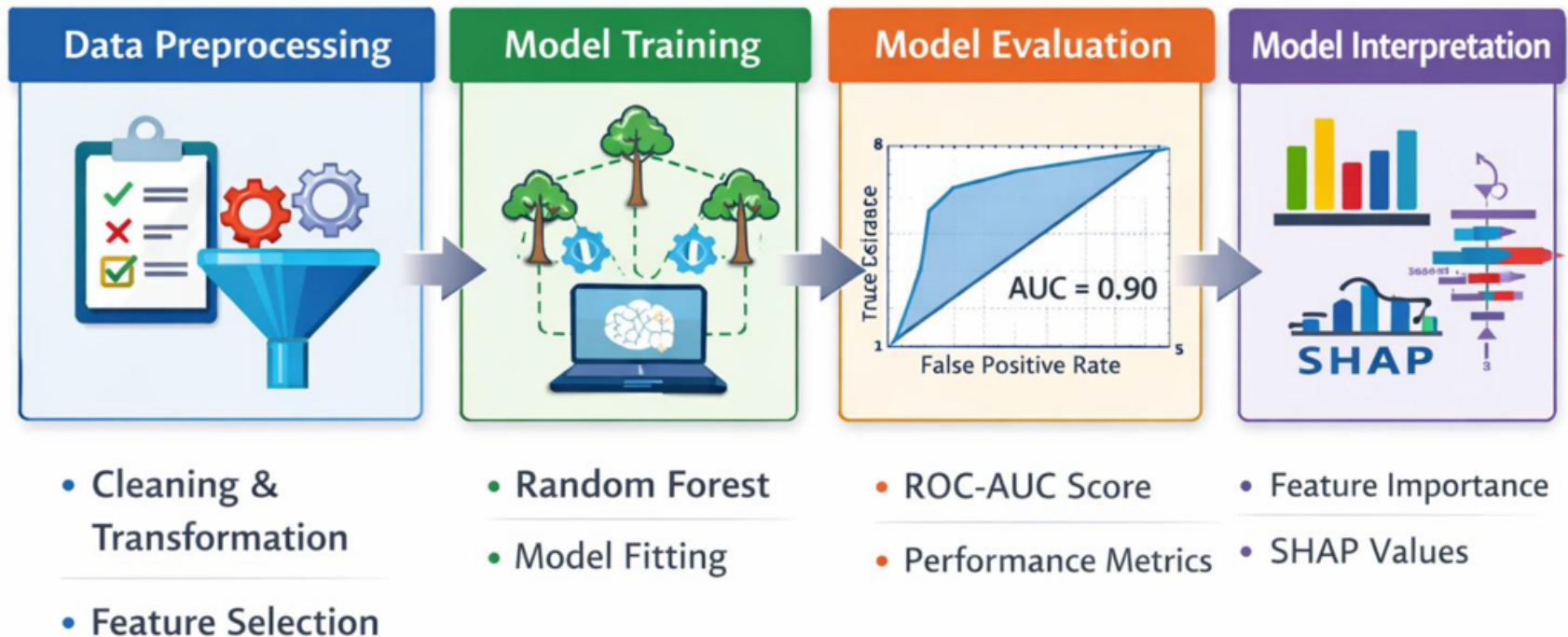
- Diabetes is a chronic disease affecting millions worldwide and requires early detection for effective management.
- Traditional diagnosis methods can be time-consuming and may not always identify risk at an early stage.
- Machine Learning provides a powerful approach to analyze medical data and predict disease risk accurately.
- This project uses advanced models like LightGBM, CatBoost, and Artificial Neural Network to predict diabetes risk.
- The system also integrates Explainable AI techniques to ensure transparency and trust in predictions.

Early detection of diabetes is challenging due to reliance on traditional diagnostic methods that are often time-consuming and dependent on manual analysis. Many existing systems lack accuracy and fail to identify high-risk patients at an early stage. Additionally, most machine learning models act as black boxes, making their predictions difficult to interpret. Therefore, there is a need for an efficient, accurate, and explainable system that can predict diabetes risk and support healthcare professionals in decision-making.

- Develop an intelligent ML-based system for diabetes prediction
- Compare multiple classification algorithms for performance
- Improve prediction accuracy using advanced models like XGBoost
- Evaluate models using multiple performance metrics
- Integrate Explainable AI for interpretability
- Identify key health parameters influencing diabetes risk
- Build a system that can support clinical decision-making

- Data preprocessing
- Handling missing values
- Replacing invalid zero values
- Feature engineering and normalization
- Splitting dataset into training and testing sets
- Applying multiple machine learning algorithms
- Evaluating models using standard metrics
- Comparing model performance using graphs
- Applying Explainable AI techniques for interpretation

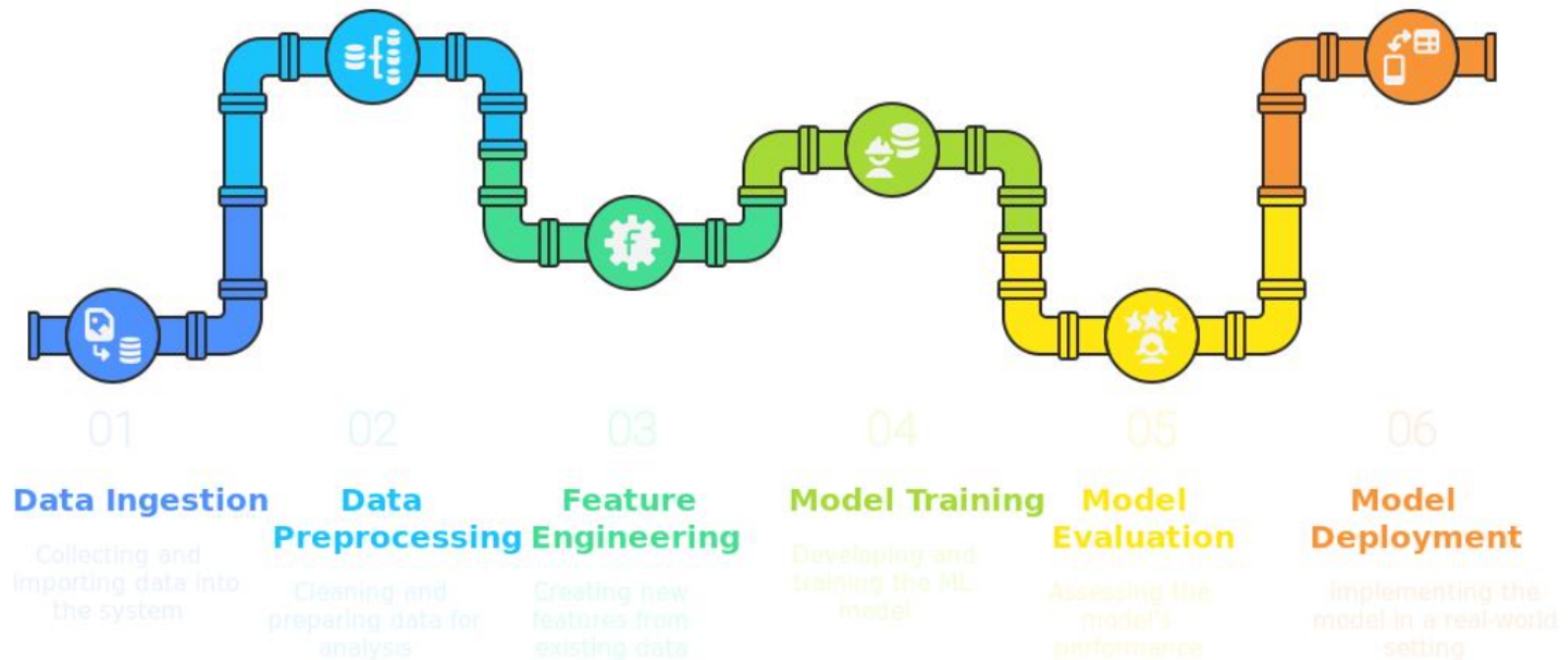
Machine Learning Pipeline for Healthcare Data



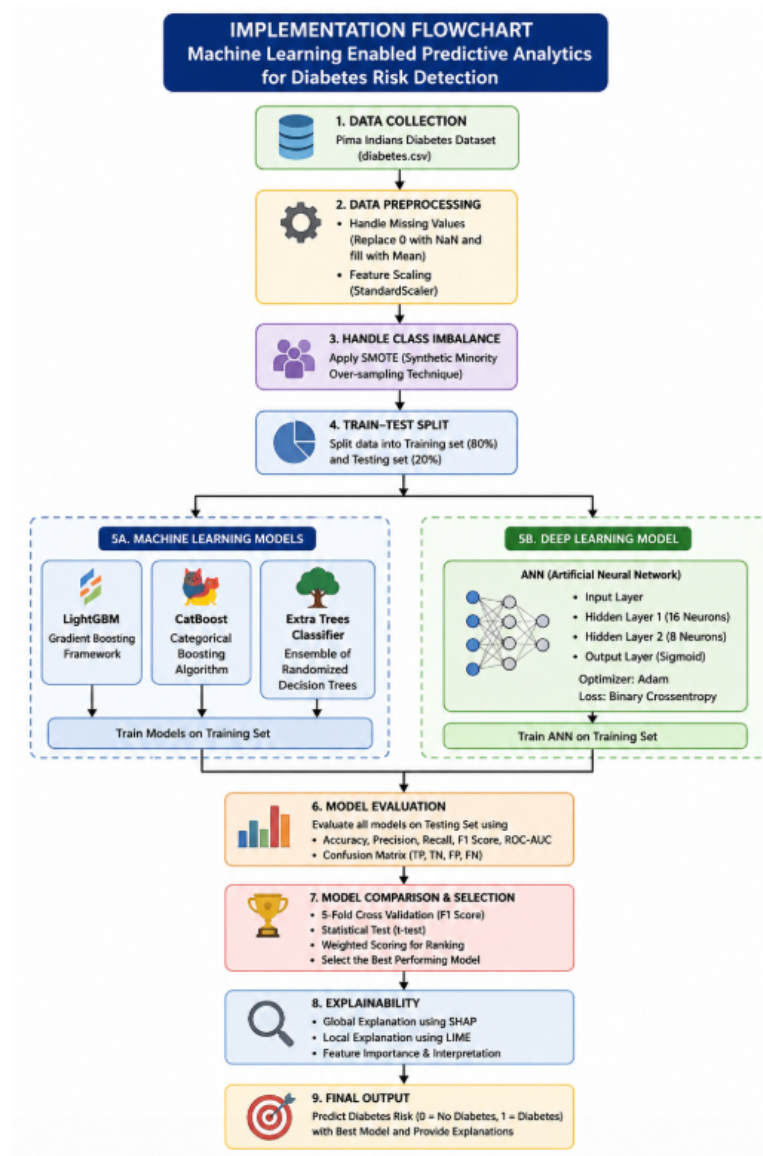
METHODOLOGY



Machine Learning Pipeline Flow

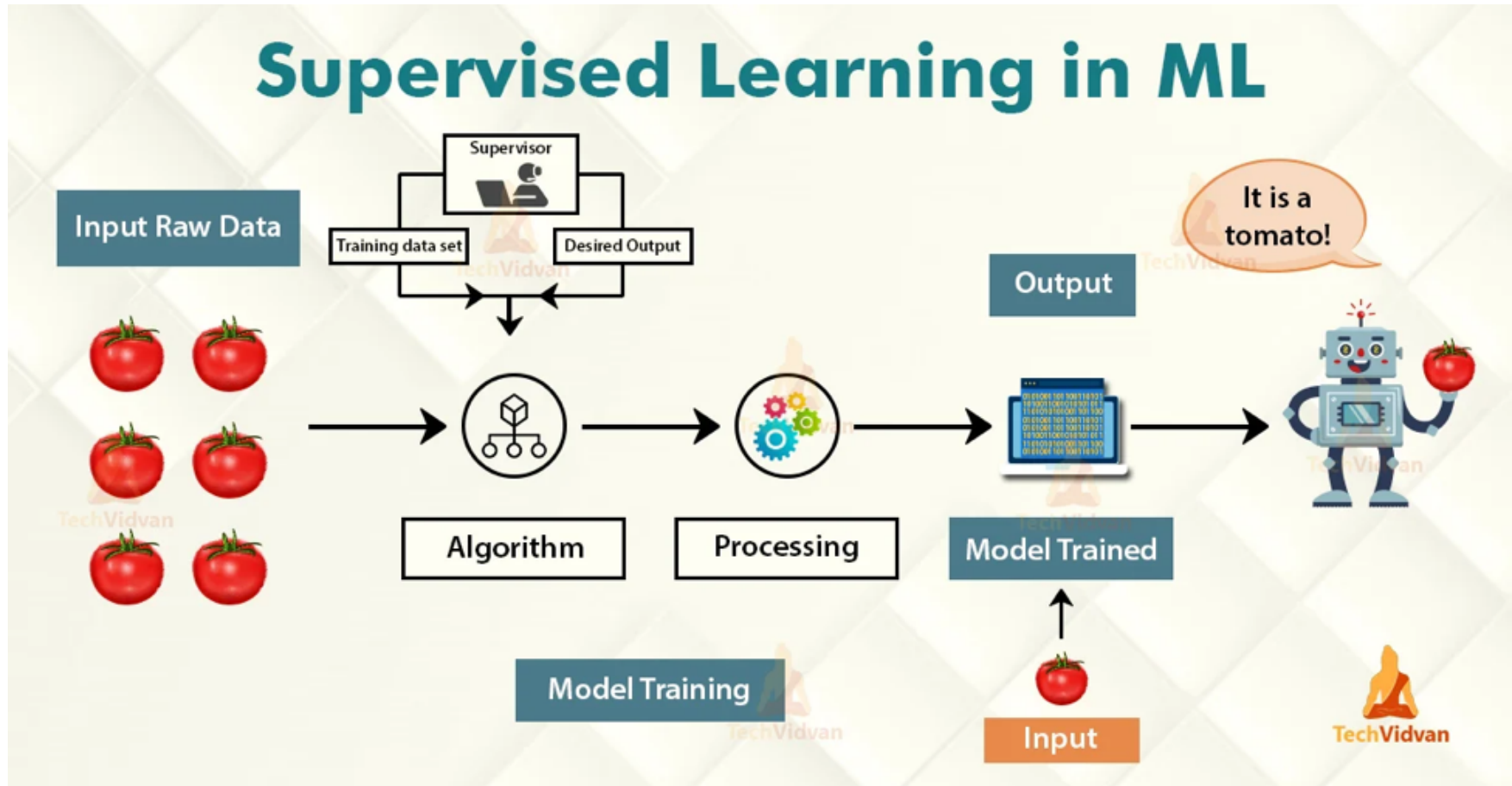


FLOWCHART



- These are the Supervised Learning ML models used-
- **Ensemble Machine Learning Models**
Used advanced ensemble models like LightGBM, CatBoost, and Extra Trees Classifier to improve prediction accuracy and robustness.
- **Deep Learning Model (ANN)**
Implemented an Artificial Neural Network with multiple dense layers to capture complex nonlinear relationships in diabetes data.
- **Handling Data Imbalance**
Applied SMOTE to balance the dataset and improve model performance on minority (diabetic) cases.
- **Model Evaluation & Selection**
Compared models using Accuracy, Precision, Recall, F1-score, and ROC-AUC, followed by weighted scoring and statistical testing to select the best-performing model.

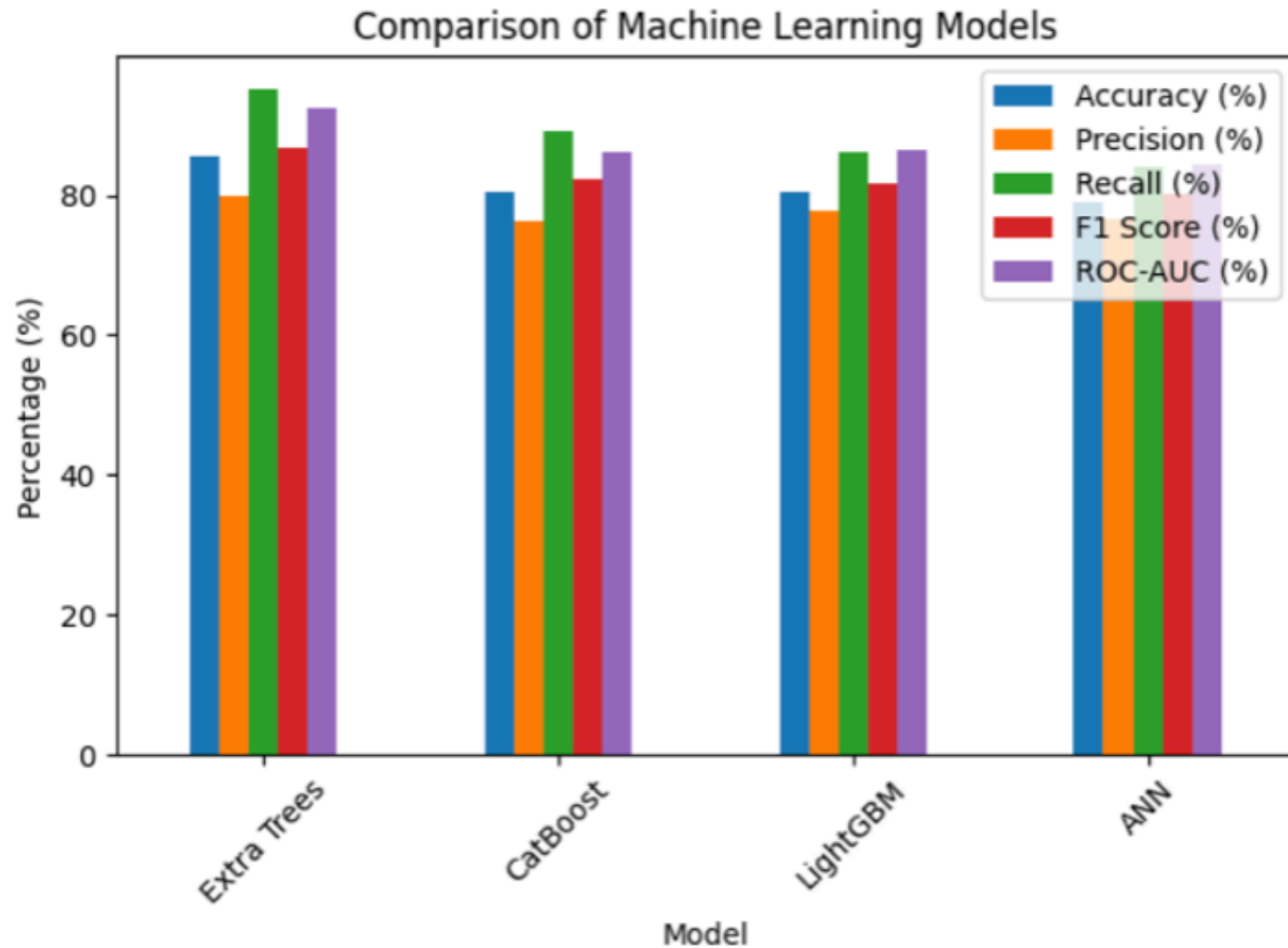
Supervised Learning in ML



- **Accuracy:** overall correctness of the model
- **Precision:** correctness of positive predictions
- **Recall:** ability to detect diabetic cases
- **F1 Score:** balance between precision and recall
- **ROC-AUC:** model's ability to distinguish classes

- XGBoost achieved highest accuracy and ROC-AUC
- Random Forest performed consistently well
- Logistic Regression showed stable baseline results
- SVM performed better after feature scaling
- Ensemble models outperformed single models
- Graphs clearly show performance differences
- ROC curve indicates strong classification ability

Results



- SHAP:Provides global feature importance
- Shows how each feature affects prediction
- LIME:Explains individual predictions
- Helps understand specific cases
- Improves transparency of ML models
- Builds trust among healthcare professionals
- Makes model decisions interpretable

- ML models can effectively predict diabetes risk
- XGBoost gives best performance among all models
- Explainable AI ensures transparency
- System is reliable and efficient
- Can assist doctors in decision-makingHelps in early detection and prevention



- Integration with hospital systems
- Development of web/mobile application
- Use of deep learning models
- Inclusion of lifestyle and genetic data
- Real-time monitoring and alerts



Thank You